4. Debt vs Equity

Two Caltech undergraduates (J&J) create a new windmill that is adaptable to extremely cold and wet climates. To develop this invention, they need 75 million dollars in capital for plant, equipment, and advances to power companies in Greenland and Nova Scotia. Net of variable costs (labor, supplies, shipping…) the firm would clear 7 million dollars a year starting two years after starting for 20 year.

A. What interest rate makes investors indifferent between starting the firm or not? Show that this is the internal rate of return of the project
B. If the interest rate is 5% and there is no equity premium show that J&J are indifferent between financing their capital needs by debt or by equity. What fraction of the firm's income value the two inventors retain after the plant gets going?
C. Now suppose there is a 20% possibility that the government of Nova Scotia decides at the end of year 1 to ban the new windmill reducing income by 50%. Show again that if markets are efficient, combinations of debt and equity do not affect the value of the firm.
D. What financial structure implies that after an adverse regulatory decision by the government of Nova Scotia, the company would belong to bondholders?
E. Suppose now that interest rates are 6% What financial structure implies that after an adverse regulatory decision by the government of Nova Scotia, the company would belong to bondholders?
F. Suppose that J&J are risk averse, what financial structure are they likely to choose? (1) if they are sure there will be no bad news, or if they have to worry about regulatory decisions.

A) IRR CALC

SOLVE USING CALC

$$\$75M = \sum_{t=2}^{21} \frac{\$7M}{(1+r)^t} \Rightarrow t = 6.12\%$$

B) PV of Annuity discounted back 1 year $A_p = \left[P \cdot \frac{1-(1+r)^{-n}}{r}\right] \frac{1}{1+r}$

$$= \left[7 \times \frac{1-(1.05)^{-20}}{0.05}\right] \frac{1}{1.05} = \$83.08M$$

NPV = \$83 − 75 MM = \$8 M belongs to J&J

J&J own $= \frac{75}{83} = 90.3\%$

C) $PV_{bad} = \sum_{t=2}^{21} \frac{\$3.5M}{(1.05)^t} = \$41.6M$

PV_{good} = same as before = \$83.08 M

Expected Value = $0.2(41.6) + 0.8(83.08) = \$74.7M \sim \$75M$

∴ IN EFFICIENT MKTS, COMBINATIONS OF DEBT & EQUITY DON'T AFFECT FIRM VALUE

D) A structure under which debt exceeds \$41.6 M would wipe out equity value

$\therefore$ Targeted debt cannot exceed $\frac{\$41.6}{75.0} = 55.4\%$ OF CAPITAL STRUCTURE

E) When interest rate ↑ from 5% to 6%

$$PV_{good} = \sum_{t=2}^{21} \frac{7}{(1.06)^t}$$

$$= 7 \cdot \left[\frac{1-(1.06)^{-20}}{0.06} \right] \cdot \frac{1}{1.06} = \$75.74\text{ M}$$

$$PV_{bad} = \frac{PV_{good}}{2} = \$37.87\text{ M}$$

$\therefore$ Expected Value $= 20\%(\$37.87) + 80\%(75.74)$
$= \$68.17$ MM

A structure under which debt exceeds \$37.87 M would wipe out equity value. $\therefore$ Targeted debt cannot exceed $\frac{\$37.87}{68.17} = 55.5\%$ (same as in part D)!

F) If J&J were to be risk averse they would choose equity as regulatory decisions and changes (increase) in interest rate can reduce Expected equity value